Supply Chain Inventory Optimization Dashboard

Primary Review Report — Final Draft

1. Case Study Title

Supply Chain Inventory Optimization Dashboard: Analyzing Stock Levels, Demand, Reorder Requirements, and Supplier Performance

Short dashboard title: **Supply Chain Inventory Optimization Dashboard**.

2. Dataset Source and Description

The selected dataset is the CSV file provided for this case study. It contains inventory and supply-chain observations at SKU and warehouse level, with sales, inventory, supplier, replenishment, pricing, promotion, and demand-forecast information.

Property	Initial finding
Records	91,250
Attributes	15
Date range	01 Jan 2024 to 30 Dec 2024
SKUs	50
Warehouses	5
Suppliers	10
Regions	4

Why this dataset is appropriate

The dataset directly contains Inventory_Level, Reorder_Point, Units_Sold, Demand_Forecast, Supplier_Lead_Time_Days, Supplier_ID, Warehouse_ID, Unit_Cost, Unit_Price, Promotion_Flag, and Stockout_Flag. These variables support operational, financial, demand, replenishment, supplier, and warehouse analysis in one dashboard.

3. Problem Statement

Effective inventory management requires organizations to maintain sufficient stock to satisfy customer demand while avoiding excessive inventory and unnecessary holding costs. Differences in demand, warehouse inventory levels, supplier lead times, reorder points, and replenishment quantities can make inventory planning difficult.

This case study proposes an interactive Supply Chain Inventory Optimization Dashboard to analyze inventory levels, sales demand, stockout indicators, reorder requirements, supplier lead times, and inventory costs across products, warehouses, suppliers, and regions. The dashboard is intended to identify potential replenishment risks, overstocking conditions, high-demand products, supplier-related issues, and opportunities for improving inventory decisions.

4. Objectives of the Analysis

Objective 1. Analyze inventory levels across SKUs, warehouses, and regions to identify relatively high and low stock positions.

Objective 2. Identify records where inventory is below or at the reorder point and prioritize potential replenishment needs.

Objective 3. Analyze units sold and demand forecasts to understand demand patterns and forecast differences.

Objective 4. Evaluate supplier lead times and compare supplier performance across warehouses and regions.

Objective 5. Estimate inventory value using inventory quantity and unit cost, then identify areas with high capital tied up in stock.

Objective 6. Examine whether promotional activity is associated with changes in sales demand and inventory pressure.

Objective 7. Provide an interactive dashboard that allows users to filter results by date, SKU, warehouse, supplier, and region.

5. Important Attributes / Data Dictionary

Attribute	Type	Role in analysis
Date	Date	Time trends and period filtering
SKU_ID	Categorical	Product/SKU-level analysis
Warehouse_ID	Categorical	Warehouse comparison
Supplier_ID	Categorical	Supplier analysis
Region	Categorical	Geographic comparison
Units_Sold	Integer	Actual demand/sales measure
Inventory_Level	Integer	Core inventory measure
Supplier_Lead_Time_Days	Integer	Replenishment/supplier risk
Reorder_Point	Integer	Replenishment threshold
Order_Quantity	Integer	Replenishment quantity

Attribute	Type	Role in analysis
Unit_Cost	Decimal	Inventory valuation
Unit_Price	Decimal	Sales/value context
Promotion_Flag	Binary	Promotion vs non-promotion comparison
Stockout_Flag	Binary	Stockout indicator; requires validation
Demand_Forecast	Decimal	Forecast vs actual demand

6. Initial Data Quality Assessment

The following findings are based on a programmatic inspection of the selected CSV and should be rechecked after cleaning and transformation.

Check	Finding	Implication / action
Missing values	0 across all 15 columns	No imputation currently required; continue monitoring after transformations.
Duplicate rows	0	No exact duplicate records detected.
Date validity	0 invalid dates; 01 Jan 2024–30 Dec 2024	Date can be converted to a proper date type.
Negative numeric values	0 in inspected numeric fields	No obvious negative-value issue.
Promotion_Flag	Values are 0/1	Suitable for binary analysis.
Stockout_Flag	Only value observed is 0	No positive stockout events are present; validate the source field.
Order_Quantity	86,223 zero records	Validate whether zero means no replenishment order.
Inventory vs reorder	4,787 below; 254 equal	Useful basis for replenishment-risk indicator.

Important finding to address before final submission

Stockout_Flag is constant at 0 in the selected file. We should not claim that the dataset contains observed stockout events. Instead, after validation, the dashboard can use a transparent replenishment-risk indicator based on Inventory_Level relative to Reorder_Point and other available variables.

7. Proposed Data Cleaning and Transformation

1. Convert Date to a proper date type and create Year, Month, Month Name, and period fields if useful.

2. Verify the business meaning and uniqueness of SKU_ID, Warehouse_ID, and Supplier_ID.

3. Validate Promotion_Flag and Stockout_Flag against the dataset documentation.

4. Investigate zero Order_Quantity values and document their meaning.

5. Check Unit_Price versus Unit_Cost for logical consistency and unusual relationships.

6. Create Inventory Value = Inventory Level × Unit Cost.

7. Create Reorder Gap = Inventory Level − Reorder Point.

8. Create Forecast Error = Units Sold − Demand Forecast and optionally absolute forecast error.

9. Create a Replenishment Risk classification based on inventory relative to reorder point; justify thresholds.

10. Recheck outliers after transformations and document any treatment.

8. Selected Visualization Tool

Tableau will be used as the primary visualization and dashboarding tool for this project. Tableau is well suited to the Supply Chain Inventory Optimization Dashboard because it supports interactive dashboards, filtering, drill-down analysis, calculated fields, KPI cards, trend analysis, and comparisons across products, warehouses, suppliers, and regions.

Python (Pandas/Matplotlib) will be used as a supporting tool for initial data profiling, data-quality checks, exploratory analysis, and validation of calculations before the final dashboard is developed in Tableau.

Tableau capabilities and planned use: Interactive filters for Date, SKU, Warehouse, Supplier, and Region; KPI cards for inventory and sales measures; calculated fields for Inventory Value, Reorder Gap, Forecast Error, and Risk Category; trend charts for time analysis; comparison charts for warehouses and suppliers; and dashboard actions for interactive exploration.

9. Proposed Visualization Plan

#	Visualization	Purpose / question
1	KPI cards	Overall inventory, sales, inventory value, lead time, and replenishment-risk position.
2	Inventory trend line	How does inventory change over time?
3	Inventory by warehouse	Which warehouses hold the most inventory?
4	Inventory vs reorder point	Which SKUs are below or near their replenishment threshold?
5	Actual sales vs demand forecast	Where does actual demand differ from forecast demand?
6	Supplier lead-time comparison	Which suppliers have longer lead times and may require closer monitoring?
7	Inventory value by warehouse/SKU	Where is the greatest amount of capital tied up in inventory?

#	Visualization	Purpose / question
8	Promotion vs units sold	Is promotional activity associated with higher demand?

10. Proposed Dashboard Layout

Header: Dashboard title and reporting period. **KPI row:** Total Inventory, Units Sold, Inventory Value, Replenishment Risk, Average Lead Time, and SKUs Below Reorder Point. **Filter row:** Date, SKU, Warehouse, Supplier, and Region. **Main analysis:** inventory trend, warehouse comparison, reorder analysis, forecast comparison, supplier lead time, and inventory-value analysis.

11. Expected Business / Analytical Insights

1. Identify SKUs and warehouses where inventory falls below the reorder point.

2. Identify locations or products with relatively high inventory and potential overstocking.

3. Determine which suppliers have the longest lead times and may require stronger replenishment planning.

4. Identify products with the largest differences between actual sales and demand forecasts.

5. Estimate where inventory capital is concentrated using inventory value.

6. Understand whether promotions are associated with increased sales demand and possible inventory pressure.

7. Support replenishment decisions using an explainable risk classification.

12. Primary Review Alignment

Rubric criterion	How this draft addresses it
Dataset Selection	Relevant supply-chain dataset with 91,250 records and direct inventory, demand, supplier, and warehouse variables.
Problem Statement	Defines the inventory-management challenge and connects it to replenishment, demand, supplier lead time, and decision-making.
Objectives	Seven measurable objectives are directly connected to the proposed dashboard analysis.
Dataset Understanding	Includes dataset dimensions plus a data dictionary identifying key attributes, types, and analytical roles.
Tools & Visualization Plan	Tableau is the primary visualization/dashboard tool, with Python used for profiling and validation; eight planned visuals are specified.
Data Quality	Documents missing values, duplicates, date validity, binary fields, zero-order records, and the constant Stockout_Flag issue.

13. Tableau Implementation Plan

The final analytical dashboard will be developed in Tableau using the selected supply-chain dataset. The implementation will follow a structured workflow so that every visualization is connected to a defined business question and every calculated measure can be explained during the project review.

13.1 Data Connection and Preparation

The CSV file will first be connected to Tableau and the data types will be checked. Date fields will be recognized as dates, identifiers such as SKU_ID and Warehouse_ID will be treated as dimensions, and numerical variables such as Inventory_Level, Units_Sold, Unit_Cost, and Demand_Forecast will be treated as measures. The data will be reviewed again after loading because Tableau's field interpretation can affect calculations and visualizations.

13.2 Proposed Calculated Fields

Calculated field	Purpose	Business interpretation
Inventory Value	Inventory_Level × Unit_Cost	Estimated capital currently held in inventory.
Reorder Gap	Inventory_Level – Reorder_Point	Negative values indicate inventory below the reorder threshold.
Forecast Error	Units_Sold – Demand_Forecast	Shows whether actual demand is above or below the forecast.
Absolute Forecast Error	ABS(Forecast Error)	Measures forecast deviation without considering direction.
Replenishment Risk	Classification using Inventory_Level and Reorder_Point	Creates an easy-to-read risk category for dashboard users.
Inventory-to-Reorder Ratio	Inventory_Level ÷ Reorder_Point	Shows inventory position relative to the replenishment threshold.

13.3 Dashboard Filters

The dashboard will include interactive filters for Date, SKU_ID, Warehouse_ID, Supplier_ID, and Region. These filters will allow users to move from an overall supply-chain view to a specific warehouse, supplier, or product. The filter design will be kept consistent across the main dashboard so that users can understand how one selection changes multiple visualizations.

13.4 Proposed Tableau Worksheets

Worksheet	Chart type	Main purpose
Inventory KPI	KPI cards	Provide a quick overview of inventory and operational measures.

Worksheet	Chart type	Main purpose
Inventory Trend	Line chart	Track inventory movement over time.
Warehouse Inventory	Bar chart	Compare inventory levels between warehouses.
Reorder Risk	Bar / highlight table	Locate SKUs below or close to reorder point.
Actual vs Forecast	Line / bar comparison	Evaluate demand forecast differences.
Supplier Lead Time	Bar chart	Compare supplier lead times.
Inventory Value	Bar chart	Identify warehouses or SKUs with high inventory value.
Promotion Analysis	Bar / comparison	Compare sales behaviour for promotion and non-promotion records.

13.5 Dashboard Interaction Strategy

Dashboard actions will be used to make the final product more than a collection of static charts. Selecting a warehouse or supplier can highlight related records in other worksheets. Tooltips will provide additional details such as SKU, inventory level, reorder point, lead time, and calculated risk without adding visual clutter.

Implementation principle: every chart should answer a specific supply-chain question. Visual elements that do not contribute to a decision or analytical objective should be removed from the final dashboard.

14. Analytical Framework, Insights and Limitations

The dashboard is intended to transform the raw inventory records into a decision-support view. The analysis will move from descriptive monitoring to diagnostic investigation: first identifying where inventory conditions differ, then examining possible relationships with demand, suppliers, warehouses, and replenishment requirements.

14.1 Key Analytical Questions

Area	Question to answer	Potential decision
Inventory position	Which SKUs or warehouses have the highest and lowest inventory levels?	Prioritize inventory review and balancing.
Replenishment	Which records are below or close to the reorder point?	Identify potential replenishment requirements.
Demand	Where is actual sales demand substantially different from forecast demand?	Review forecasting assumptions.
Suppliers	Which suppliers have longer lead times?	Consider supplier monitoring and planning buffers.

Area	Question to answer	Potential decision
Inventory value	Where is the greatest capital tied up in inventory?	Investigate high-value stock and working-capital exposure.
Promotions	How do units sold differ between promotion and non-promotion records?	Evaluate whether inventory planning should account for promotion periods.
Geography	How do regions differ in inventory and demand measures?	Support regional planning and allocation.

14.2 Expected Analytical Insights

The final dashboard should reveal which parts of the supply chain require attention rather than simply reporting totals. For example, a warehouse with high inventory may appear healthy from a stock-availability perspective but may also have a high inventory value. Conversely, a warehouse with low inventory may carry greater replenishment risk if its reorder gap is negative and supplier lead time is relatively long.

Comparing Units_Sold with Demand_Forecast can identify products for which demand is consistently higher or lower than expected. Combining this information with inventory levels can help distinguish potential demand pressure from simple excess-stock situations. Supplier lead-time analysis can then provide additional operational context.

14.3 Data Limitations and Responsible Interpretation

The selected dataset has several characteristics that must be acknowledged in the final presentation. The Stockout_Flag field contains only zero values in the current file, so the project should not claim that actual stockout events were observed. Instead, replenishment risk should be derived transparently from the available inventory and reorder information.

Relationships shown in the dashboard should be interpreted as patterns in the available data, not automatic proof of causation. For example, if promotion records show higher units sold, this indicates an association in the dataset; it does not by itself prove that promotion caused the increase.

14.4 Final Dashboard Success Criteria

1. Clear hierarchy: users should understand the overall inventory position within a few seconds.

2. Consistency: the same filters and definitions should be used across related visualizations.

3. Actionability: each major chart should support a business question or decision.

4. Transparency: calculated fields and risk classifications should be explainable.

5. Accuracy: dashboard totals should be cross-checked against the source dataset and validation results.

6. Usability: labels, tooltips, legends, and units should be readable without overcrowding the dashboard.

7. Rubric alignment: the final Tableau dashboard should visibly connect the dataset, problem statement, objectives, analysis, and expected business insights.

Overall project direction: Tableau will serve as the final interactive presentation layer, while Python will support data validation and exploratory analysis. The completed dashboard will combine inventory, demand, replenishment, supplier, warehouse, and financial perspectives into one coherent decision-support tool.